

Queuing Models / Waiting Line Model

Why is there waiting?

- ✓ Waiting lines occur when there is a temporary imbalance between supply (service) and demand (arrival of customers).
- ✓ Because services cannot be performed ahead of time and stored until needed, the system at times becomes temporarily overloaded, giving rise to waiting lines.

From a managerial perspective, the key is to determine the balance that will provide an adequate level of service at a reasonable cost.

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Characteristic of Waiting line

The main characteristics are:

1. Population source
2. Number of servers (channels)
3. Arrival and service patterns
4. Queue discipline (order of service)
5. Queue length



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Characteristic of Waiting line

1. Population source

- ✓ **Finite-Source:** When the potential number of customers is limited.

e.g., If a repairman responsible for 10 machines in a company. The potential number of machines that might need repairs at any one time cannot exceed 10.

- ✓ **Infinite-source:** When the potential number of customers is unlimited.

e.g., Supermarkets, banks, and toll bridges.

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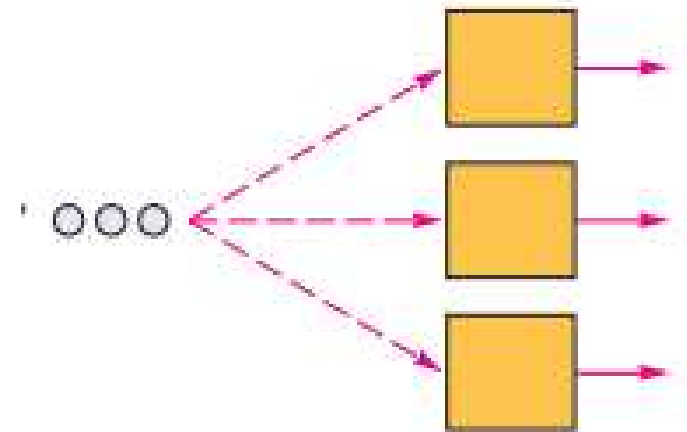
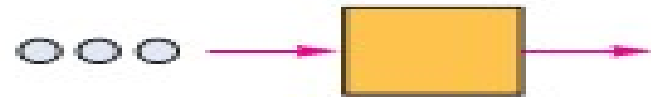
Characteristic of Waiting line

2. Number of servers (channels)

- ✓ Systems can be either **single-** or **multiple-server**.

Examples of **single-server systems** are small grocery stores with one checkout counter, single-bay car washes.

Multiple-server systems are commonly found in banks, at airline ticket counters, railway reservation counter.

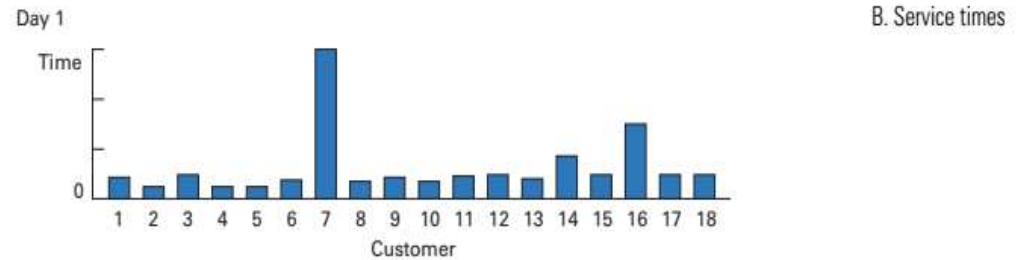
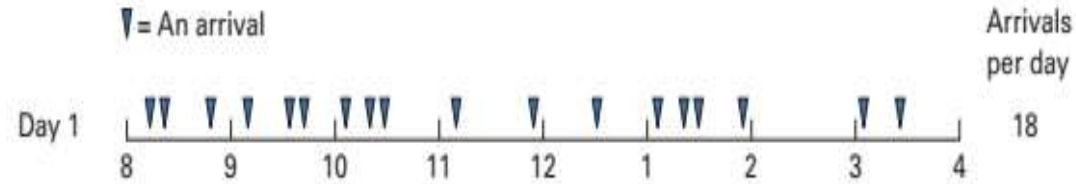


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3. Arrival and service patterns

- ✓ Waiting lines are a direct result of arrival and service variability.
- ✓ The variabilities can be described by theoretical distributions.
- ✓ Assumption is that the arrival can be described by a Poisson distribution and the service time can be described by an Exponential distribution.



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Characteristic of Waiting line

4. Queue discipline (order of service)

- ✓ Queue discipline refers to the order in which customers are processed

FCFS (FIFO), e.g., Banks, stores, post-office

Priority, e.g., hospital emergency

5. Queue length

Finite queue length, e.g., car service centre (due to parking limitation)

Infinite queue length, e.g., Reservation counter

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Reneging, Jockeying, & Balking

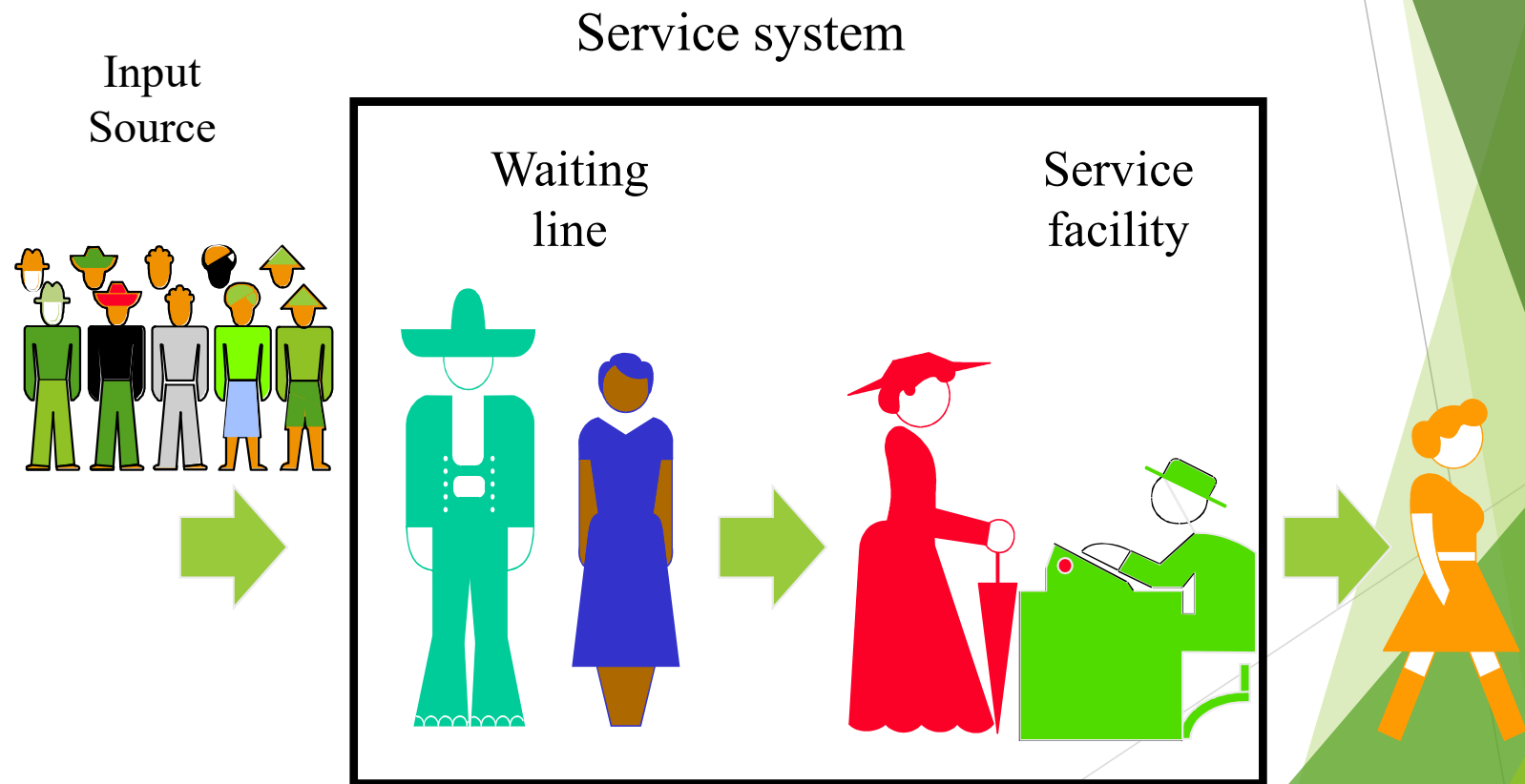
✓ When customer become impatient Reneging, Jockeying, & Balking occur.

1. Reneging: waiting customers grow impatient and leave the line
2. Jockeying: customers switch to another line
3. Balking: upon arriving, customers decide the line is too long and, therefore, do not enter the line.

Note: The models in this chapter assume customers are patient, that is, that customers enter the waiting line and remain until they are served.

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Queuing System/ Waiting Line System



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Measures of Waiting-line performance

The operations manager typically looks at three measures when evaluating existing or proposed service systems.

1. The average number of customers waiting, either in line or in the system.
2. The average time customers wait, either in line or in the system.
3. System utilization, which refers to the percentage of capacity utilized.

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Symbol	Represents
λ	Customer arrival rate
μ	Service rate per server
L_q	The average number of customers waiting for service
L_s	The average number of customers in the system (waiting and/or being served)
r	The average number of customers being served
ρ	The system utilization
W_q	The average time customers wait in line
W_s	The average time customers spend in the system (waiting in line and service time)
$1/\mu$	Service time

Note: The arrival and service rates, represented by λ and μ , must be in the same units (e.g., customers per hour, customers per minute).

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- ✓ The average number of customers being served

$$r = \frac{\lambda}{\mu}$$

- ✓ The average number of customers waiting in line

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

- ✓ The average number of customers waiting in the system

$$L_s = L_q + r$$

- ✓ The average time customers wait in $W_q = \frac{L_q}{\lambda}$

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- ✓ The average time customers wait in system

$$W_s = W_q + \frac{1}{\mu} = \frac{L_s}{\lambda}$$

- ✓ System utilization $\rho = \frac{\lambda}{M\mu}$

- ✓ Probability of zero customer in the system

$$P_0 = 1 - \left(\frac{\lambda}{\mu} \right)$$

- ✓ Probability of n customers in the system

$$P_n = P_0 \left(\frac{\lambda}{\mu} \right)^n$$

- ✓ Probability of less than n customers in the system

$$P_{<n} = 1 - \left(\frac{\lambda}{\mu} \right)^n$$

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Q. Customers arrive at a bakery at an average rate of 18 per hour on weekday mornings. The arrival distribution can be described by a Poisson distribution with a mean of 18. Each clerk can serve a customer in an average of three minutes; this time can be described by an exponential distribution with a mean of 3.0 minutes.

- What are the arrival and service *rates*?
- Compute the average number of customers being served at any time.
- Suppose the average number of customers waiting in line is 8.1. Compute the average number of customers in the system
- Determine the system utilization for $M = 1$ server.

$\lambda = 18$ customers per hour.

$\mu = 20$ customers per hour

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b. $r = \frac{\lambda}{\mu} = \frac{18}{20} = .90$ customer.

c. Given: $L_q = 8.1$ customers.

$$L_s = L_q + r = 8.1 + .90 = 9.0 \text{ customers}$$

$$W_q = \frac{L_q}{\lambda} = \frac{8.1}{18} = .45 \text{ hour}$$

$$W_s = \text{Waiting in line plus service}$$

$$= W_q + \frac{1}{\mu} = .45 + \frac{1}{20} = .50 \text{ hour}$$

d. System utilization is $\rho = \frac{\lambda}{M\mu}$.

$$\text{For } M = 1, \rho = \frac{18}{1(20)} = .90$$

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Q An airline is planning to open a ticket desk, staffed by one ticket agent. It is estimated that requests for tickets will average 15 per hour, and requests will have a Poisson distribution. Service time is assumed to be exponentially distributed. Previous experience suggests that mean service time should average about 3 minutes per request. Determine each of the following:

- a. System utilization.
- b. Percentage of time the server (agent) will be idle.
- c. The expected number of customers waiting to be served.
- d. The average time customers will spend in the system.
- e. The probability of zero customers in the system and the probability of four customers in the system.

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$\lambda = 15$ customers per hour

$\mu = 20$ customers per hour

a. $\rho = \frac{\lambda}{M\mu} = \frac{15}{1(20)} = .75$

b. Percentage idle time $= 1 - \rho = 1 - .75 = .25$, or 25 percent

c. $L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{15^2}{20(20 - 15)} = 2.25$ customers

d. $W_s = \frac{L_q}{\lambda} + \frac{1}{\mu} = \frac{2.25}{15} + \frac{1}{20} = .20$ hour, or 12 minutes

e. $P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{15}{20} = .25$ and $P_4 = P_0 \left(\frac{\lambda}{\mu} \right)^4 = .25 \left(\frac{15}{20} \right)^4 = .079$

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Kendall's notation = Model $M/M/1:\infty/\infty$ (Markovian arrival/Markovian service/single server: infinite queue length/infinite population)

Q. Assume a drive-up window at a fast-food restaurant.
Customers arrive at the rate of 25 per hour.
The employee can serve one customer every two minutes.
Assume Poisson arrival and exponential service rates

Determine:

- A) What is the average utilization of the employee? (system utilization)
- B) What is the average number of customers in line?
- C) What is the average number of customers in the system?
- D) What is the average waiting time in line?
- E) What is the average waiting time in the system?
- F) What is the probability that exactly two cars will be in the system?

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A) What is the average utilization of the employee?

$$\lambda = 25 \text{ cust / hr}$$

$$\mu = \frac{1 \text{ customer}}{2 \text{ mins (1hr / 60 mins)}} = 30 \text{ cust / hr}$$

$$\rho = \frac{\lambda}{\mu} = \frac{25 \text{ cust / hr}}{30 \text{ cust / hr}} = \mathbf{.8333}$$

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B) What is the average number of customers in line?

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{(25)^2}{30(30 - 25)} = \mathbf{4.167}$$

C) What is the average number of customers in the system?

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{25}{(30 - 25)} = \mathbf{5}$$

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D) What is the average waiting time in line?

$$W_q = \frac{L_q}{\lambda} = .1667 \text{ hrs} = \mathbf{10 \text{ mins}}$$

E) What is the average waiting time in the system?

$$W_s = \frac{L_s}{\lambda} = .2 \text{ hrs} = \mathbf{12 \text{ mins}}$$

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F) What is the probability that exactly two cars will be in the system (one being served and the other waiting in line)?

$$p_n = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n$$

$$p_2 = \left(1 - \frac{25}{30}\right) \left(\frac{25}{30}\right)^2 = .1157$$

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Q. Customers arrive at a clinic at the rate of 8/hour (Poisson arrival) and the doctor can serve at the rate of 9/hour (exponential).

- (a) What is the probability that a customer does not join the queue and walks into the doctor's room?
- (b) What is the probability that there is no queue?
- (c) What is the probability that there are 10 customers in the system?
- (d) What is the expected number in the system?
- (e) What is the expected waiting time in the queue?

Solution: Given $\lambda = 8/\text{hour}$, $\mu = 9/\text{hour}$

(a) The customer goes directly to the doctor when there is no one in the system. The probability is:

$$p_0 = 1 - \rho = 1 - \frac{8}{9} = 0.111$$

(b) There is no queue when there is either no one in the system or there is one person in the system. The probability is:

$$\begin{aligned}\text{Probability} &= p_0 + p_1 \\ &= 0.111 + 0.111 \times \frac{8}{9} = 0.2098\end{aligned}$$

(c) The probability that there are 10 customers in the system is:

$$\begin{aligned}p_{10} &= \rho^{10} p_0 \\ &= \left(\frac{8}{9}\right)^{10} \times 0.111 = 0.0341\end{aligned}$$